FEEL BEFORE YOU BUILD

Hands-on prototyping with actuators · 2 hours

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MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

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TLDR

Two hours finding out what actuators feel like, and leaving able to drive one from your own code.

Get a board and run task 01. Something spins.

Change two numbers, upload again. That loop is the workshop.

Pick up the actuators you are not driving.

Work through as many of the five tasks as you get to.

Build one signal someone else can read without looking.

WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Most projects end up on the screen and the speaker

- Not because touch was wrong
- Because nobody knew a coin motor costs 12 kr

Today: decide it feels cheap, pick something else

- Better now than in week 46

HOW TODAY WORKS

A board each, five tasks, everything pre-wired

- Upload, change two numbers, upload again
- You wire nothing today

Actuators are on the tables at the front

- Go and feel the ones your task does not use

Finish by building one signal

- Someone else has to read it

RUN SHEET

Times are offsets from the start. The only hard checkpoint is 0:25: everyone has task 01 running.

0:00	10 min	Two motors, same message. Which felt urgent?
0:10	15 min	Boards out, task 01 running for everyone
0:25	60 min	Tasks 02 to 05, at your own pace
1:25	10 min	Round-the-room: what surprised you
1:35	20 min	Blind test on task 05
1:55	5 min	Pack down

FIVE TASKS

FIVE TASKS, ONE BOARD EACH

Upload, change two numbers, upload again. That loop is the point.

01	**Make something spin**	**coin motor**
02	**Make something tap**	**solenoid**
03	**Make something warm**	**Peltier tile**
04	**Make it notice you**	**a wire and foil**
05	**Make it answer back**	**both together**
06	**Use what you have**	**your own phone**

1 to 3 are output. 4 is input. 5 is both. 6 needs no board.

Nobody is expected to finish all five.

HOW THE TASK SKETCHES WORK

Every sketch runs the moment you upload it. Nothing to wire.

A CHANGE ME block at the top

- Two or three numbers. Edit, upload, feel the difference

A THINGS TO TRY list at the bottom

- Four experiments, roughly in order of difficulty
- The last one is usually the interesting one

github.com/labtools-au/multimodal-actuator-workshop

WHAT YOU GET IN THE BASE KIT

One set per pair. Collect it before you start task 01.

Board, breadboard, jumper wires, USB cable

- Assorted resistors and a few transistors

Breadboard power supply, drawer 8A

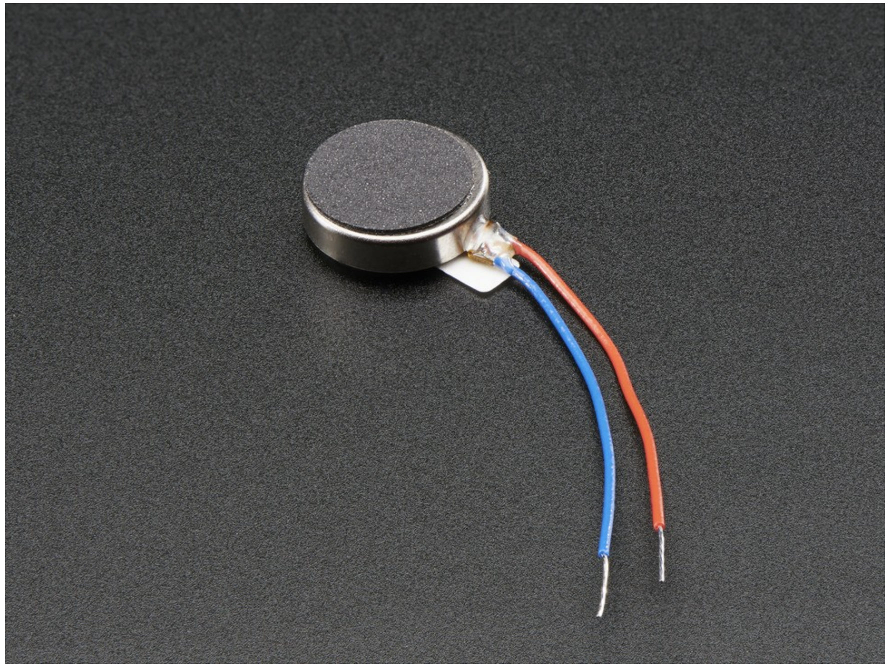
- 42 in stock. Use it for anything that is not just an LED.

L9110S H-bridge, drawer 2C

- 24 in stock. Needed for the Peltier and for reversing a motor.

WHAT IS ON THE TABLES

VIBRATING MINI MOTOR DISC



An off-centre weight on a motor shaft, sealed in a disc. The same part as in your phone.

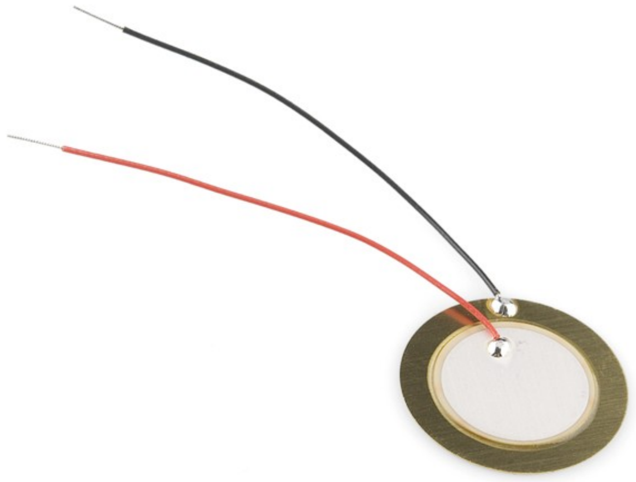
Feels like: a buzz you cannot make gentle and fast at once.

USED FOR

Phone notifications, controller rumble, anything worn under clothing.

`Drawer 1E  |  117 in stock  |  2.5-3.8 V  |  PWM pin`

PIEZO ELEMENT



A crystal that flexes when you put current across it. Near-instant, almost no power, but shallow.

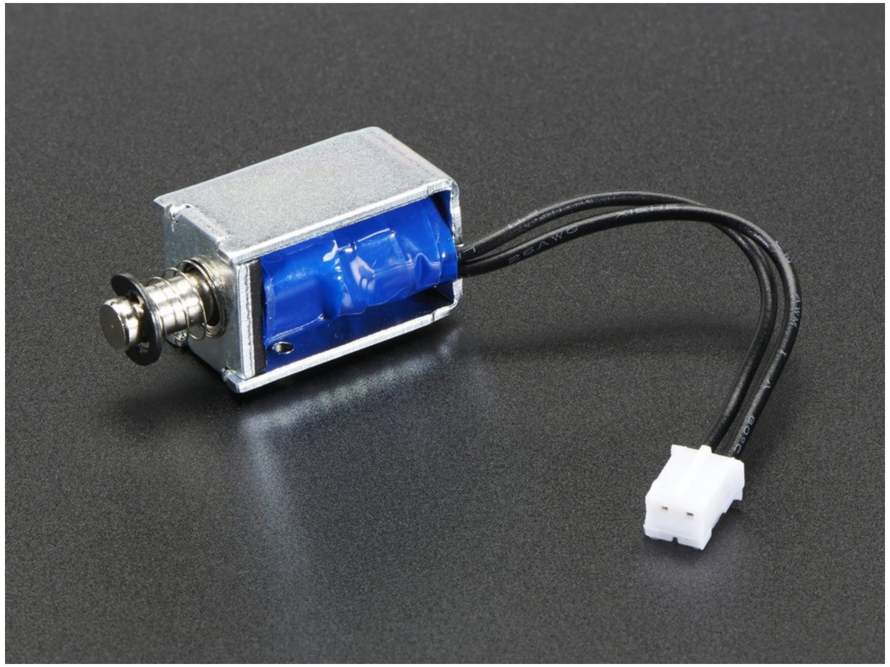
Feels like: a tick, not a thump.

USED FOR

Clicks and confirmations. Anything where lag would feel broken.

`Drawer 6F  |  69 in stock  |  any digital pin`

MINI PUSH-PULL SOLENOID



A coil that yanks a metal rod when you energise it. No half a tap: it fires or it does not.

Feels like: a person tapping you, not a machine signalling.

USED FOR

Navigation cues on the body, braille cells, alerts in noise.

`Drawer 3E  |  24 in stock  |  5V  |  needs a MOSFET`

CAPACITIVE PAD

PHOTO
Capacitive pad

A wire taped behind foil, card or fabric. No sensor, no breakout board.

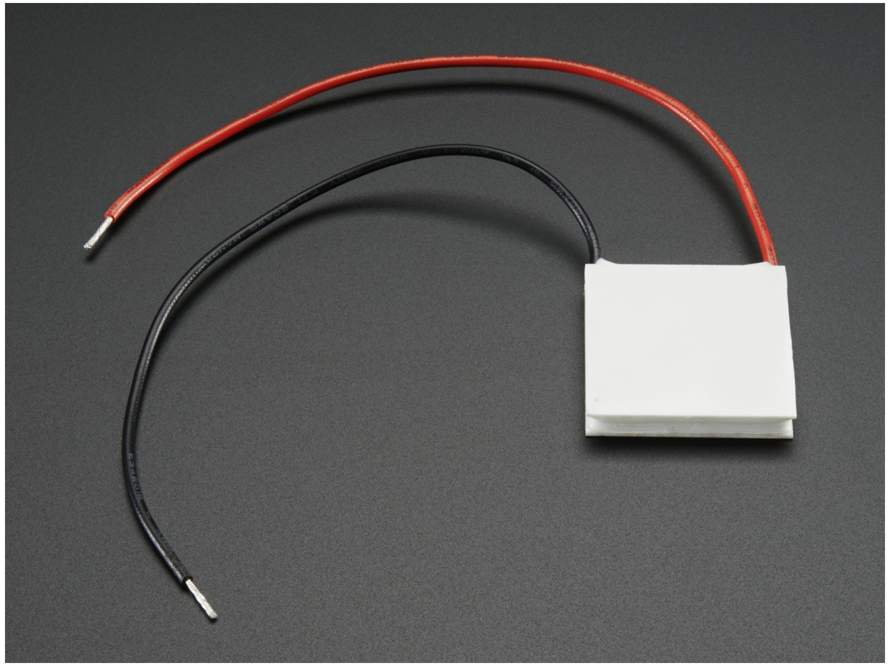
Feels like: nothing. That is the point. The surface stays plain.

USED FOR

Invisible controls in wood or fabric, waterproof panels.

`No part number. A wire, on any analogue pin`

PELTIER TILE



Heats one face and cools the other. Flip the current and it reverses.

Feels like: slow. Several seconds before you are sure which way.

USED FOR

Slow ambient state. Never an alert.

`Drawer 4C | 25 small, 35 big | needs an H bridge`

SURFACE TRANSDUCER

Turns any surface into a speaker. Drive it so one frequency is felt and another is heard, from the same part.

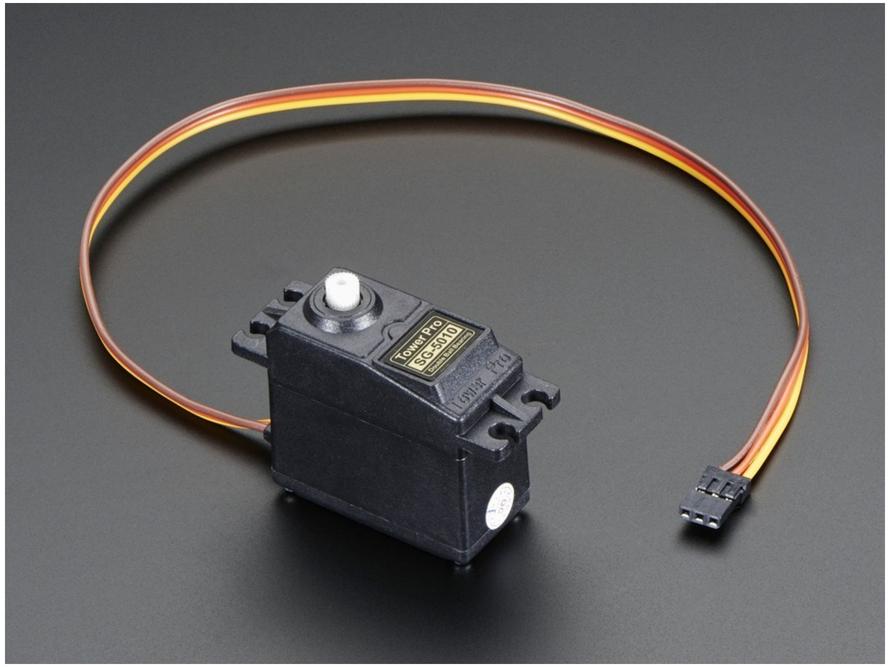
Feels like: more certain, rather than louder.

USED FOR

Noisy or bright environments, and accessibility.

`Drawer 7F  |  12 large, 13 medium, 23 bone  |  needs an amp`

SERVO



A motor that holds a position instead of spinning freely. Tell it an angle and it goes there.

Feels like: something deliberate moving, with force behind it.

USED FOR

Anything that points, pushes, opens or resists a hand.

`Drawer 2F | 80 in stock | Servo.h | needs its own 5V`

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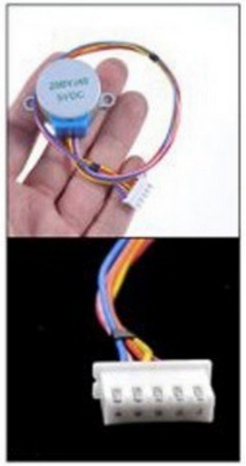
STEPPER MOTOR



Moves in fixed steps rather than continuously, so you always know where it is without a sensor.

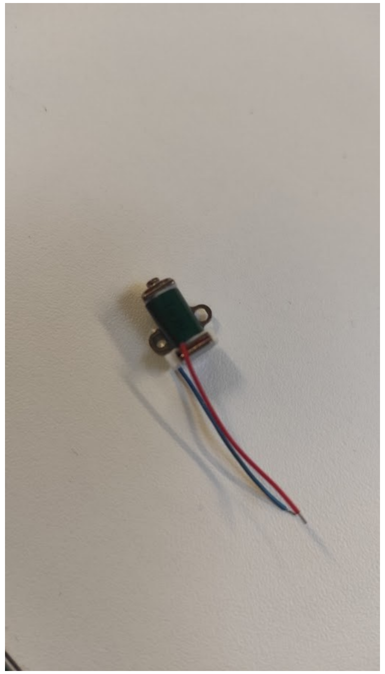
Feels like: precise, and audibly clicky.

USED FOR

Slow accurate motion. Dials, sliders, anything positioned.

`Drawer 2A  |  44 in stock  |  needs a ULN2003 driver`

ELECTROMAGNET



Grabs and releases ferrous metal on command. No moving parts of its own.

Feels like: a grip that appears and vanishes.

USED FOR

Latches, holds, and anything that should let go on cue.

`Drawer 3D | 17 mini, 11 standard | needs a MOSFET`

FREESTYLE IS ENCOURAGED

You do not have to do the tasks in order.

You do not have to finish them.

Getting task 01 working and then properly playing with task 02 beats rushing through all five.

If you want to drive something that is not in front of you, go and get it.

BEFORE YOU POWER ANYTHING

The solenoid and the Peltier both pull real current.

• They need their own supply. Ask before you rewire either.

The Peltier needs its heatsink on. Every time.

• Without it the tile cooks itself in about a minute.

Never drive a motor straight from a pin.

• 40 mA limit. The motor wants nearly twice that.

THE ONE THAT SURPRISES PEOPLE

Touch input for the price of a wire. The thinking is where it costs you.

reading > baselineAverage() * 1.01

Threshold is relative, never absolute

- Readings drift with humidity and mains hum
- Fixed threshold works at 9am, fails at 1pm

Baseline only learns while untouched

- Or a long press teaches it that a finger is normal

INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table with a sign. Wander over between tasks.

- Capacitive touch. A wire. That is task 04
- Heart rate (PPG). Useless once the hand moves
- Skin conductance (GSR). Relative change only
- Flex sensor. The cheap data glove
- Pressure (FSR). Calibrate every pad
- Your own phone. No soldering at all

THE SENSORS ALREADY IN THE ROOM

Every phone here is a multimodal device. Some of it the browser will hand you, some of it it will not.

Works in a web page today

- Accelerometer and gyroscope. Tap to grant, HTTPS only
- Microphone. Easiest continuous input there is
- Camera. Even reads a pulse off a face, badly
- Trackpad pressure. Pointer Events carry a 0 to 1 force

Does NOT work, whatever the tutorial says

- Phone vibration on iOS. WebKit has never shipped it
- Apple Watch or Garmin, live. Both need a native app

THE ONE TO CHECK BEFORE YOU PROMISE IT

navigator.vibrate: Android yes, iOS no, at every version to date.

caniuse lists Safari 3.1 to 27 as unsupported.

Chrome and Firefox on iOS are Safari underneath, so they inherit the gap.

"The phone buzzes" therefore works on some of the room and no iPhones.

Check the feature, not the tutorial, before it is in your plan.

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes

BUILD ONE SIGNAL

THE BRIEF

30 minutes. Not long enough to be precious about it.

One actuator. Three messages

- Arrived. Something wrong. Finished

Vary two things only

- Rhythm of the pulses
- How strong they are

Blind test: they name all three

- Note which two got confused, and why

THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,

you built a visual interface with a motor glued to it.

TEST IT

Hand your device to another group with the screen turned away.

- They can tell all three messages apart.
- It does not fire continuously when held.
- You know which two got confused, and what would have separated them.
- It still works when they are not looking at it.

WHAT YOU ACTUALLY LEARN

Three things you cannot get from a datasheet.

What things feel like

• A motor is a nudge. A solenoid is a knock. A Peltier is slow.

How to drive one

• PWM, direction, debouncing, and why a motor never goes straight to a pin.

What to pick

• Choosing the actuator before designing the interaction, not in week 46.

GO AND TOUCH THINGS

github.com/labtools-au/multimodal-actuator-workshop